Analysis Report: Multi-Asset Return Prediction Model

Project: S26 IC Quantitative Sector Mentorship Final Project
Objective: Implementation of a Deep Learning Pipeline for Multi-Asset Time-Series Forecasting

1. Executive Summary

This project establishes a comprehensive framework designed to predict next-day (t+1) log returns for a diversified multi-asset universe, including Equities (SPY), Fixed Income (TLT), Gold (GLD), and Energy (XLE). Inspired by the benchmark study of Oxford's Machine Learning Research Group, the architecture integrates a Variable Selection Network (VSN) with Long Short-Term Memory (LSTM) networks to capture non-linear temporal dependencies while dynamically weighting the most relevant features. The framework prioritizes analytical rigor by implementing strict anti-leakage protocols such as embargoed cross-validation and causal scaling to ensure the statistical validity of the results.

2. Data Acquisition and Preprocessing

The foundation of the framework relies on a multi-source data acquisition strategy. Financial time-series data (OHLCV) for the asset universe were retrieved via the Refinitiv Data API, while macroeconomic indicators were sourced from the Federal Reserve Economic Data (FRED).

2.1. Feature Universe

A total of 45+ features were engineered to provide the model with diverse market signals:

- **Macroeconomic Drivers:** Term Spread (10Y-2Y), Credit Spread (BAA-AAA), 10-Year Breakeven Inflation, VIX Index, and WTI Crude Oil prices.
- **Technical Indicators:** Exponential Moving Averages (EMA), MACD (12, 26), Relative Strength Index (RSI-14), and Rolling Historical Volatility (20-day).
- **Price Dynamics:** 1-day, 5-day, and 20-day lagged log returns to capture momentum and mean-reversion patterns.

2.2. Statistical Conditioning

To ensure stationarity, level-based macroeconomic data were transformed using first-order differencing, while price data were converted to log returns. A critical component of the

preprocessing is the Causal Expanding Window Scaler. Unlike standard scalers that use the entire dataset statistics, this implementation computes the mean and standard deviation only from the available historical data at each point in time, thereby eliminating look-ahead bias.

3. Feature Selection

Financial data is inherently noisy and often contains highly correlated features. To mitigate the "curse of dimensionality" and improve model interpretability, a two-stage feature selection pipeline was implemented:

Stage	Methodology	Primary Objective
1. Redundancy Filter	Hierarchical Clustering (Spearman Correlation)	Identify and group collinear variables, selecting one representative from each cluster.
2. Relevance Filter	XGBoost Feature Importance	Rank variables by their predictive contribution and select the top K (e.g., 12) features per asset.

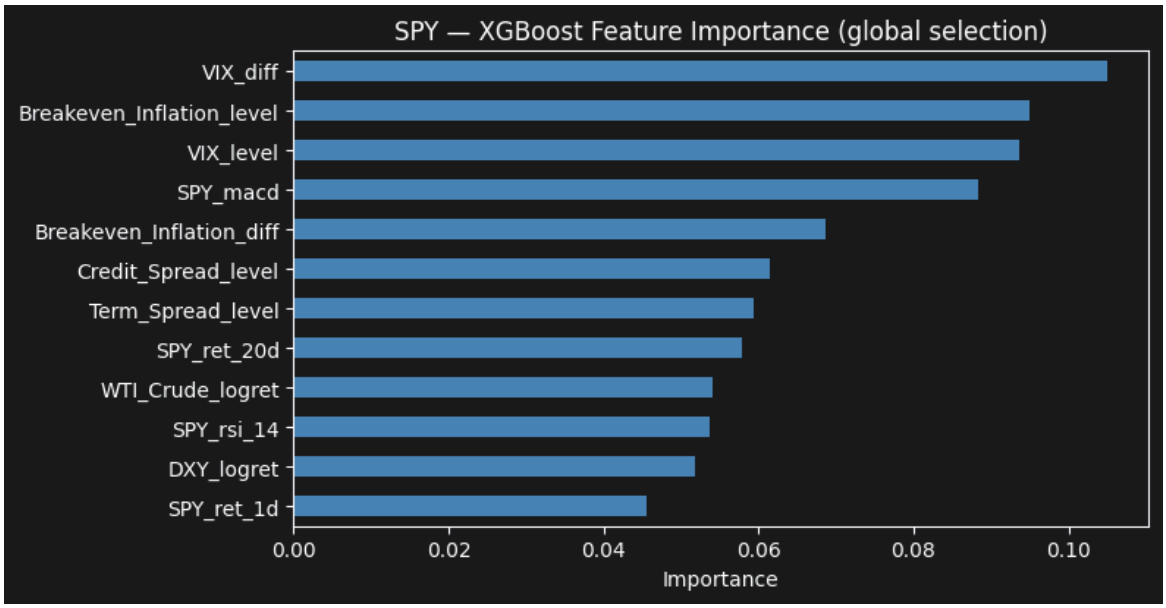

Figure 1: Feature Selection for SPY

4. Model Architecture: VSN + LSTM

The core of the predictive engine is a hybrid deep learning architecture that combines local feature selection with global temporal modeling.

4.1. Variable Selection Network (VSN)

As proposed in contemporary sequence models, the VSN uses Gated Residual Networks (GRN) to process each feature independently. It produces a set of weights that quantify the importance of each variable at every time step. This allows the model to "ignore" noise and focus on macro drivers during high-volatility regimes or technical signals during trending periods.

4.2. Temporal Learning with LSTM

The weighted feature vectors are passed into an LSTM layer with a 20-day lookback window. This enables the framework to learn complex dependencies across time, which traditional linear models often fail to capture. The model is optimized using “Huber Loss”, providing robustness against outliers frequently encountered in financial return distributions.

5. Validation Framework and Results

To simulate a real-world trading environment, the project employs an Expanding Window Cross-Validation scheme. Crucially, a 20-day Embargo is enforced between the training and test sets to prevent leakage from serial correlation in the features.

5.1. Performance Metrics

Hyperparameters were optimized individually for each asset using the Optuna framework. Key performance results are summarized below:

Asset Class	Hit Ratio (Directional Accuracy)	Root Mean Squared Error (RMSE)
Equities (SPY)	~53.3%	0.0107
Gold (GLD)	~52.2%	0.0094

Asset Class	Hit Ratio (Directional Accuracy)	Root Mean Squared Error (RMSE)
Fixed Income (TLT)	~51.4%	0.0104
Energy (XLE)	~50.4%	0.0171

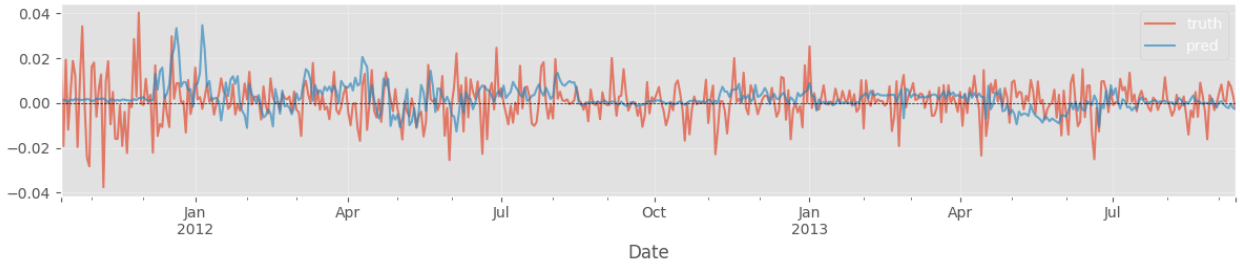

Figure 2: SPY Prediction vs Actual

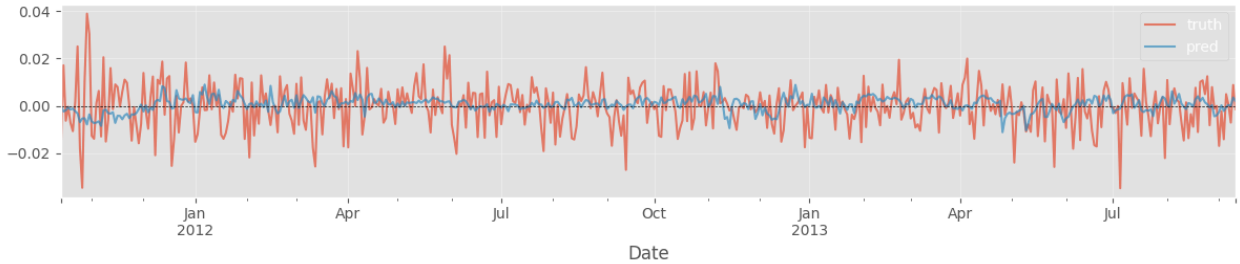

Figure 3: TLT Prediction vs Actual

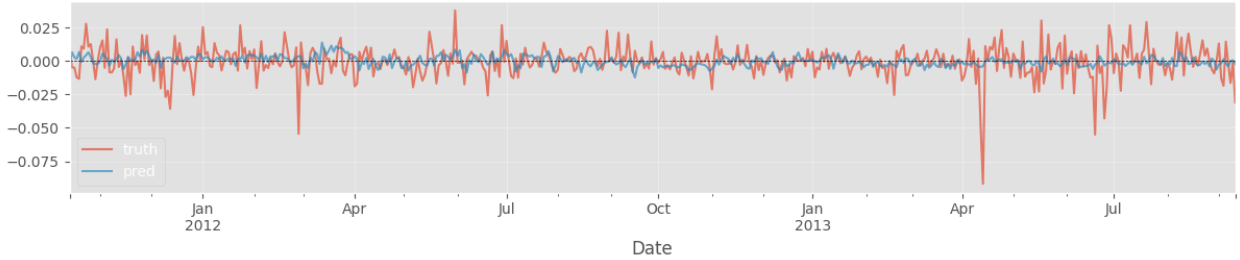

Figure 4: GLD Prediction vs Actual

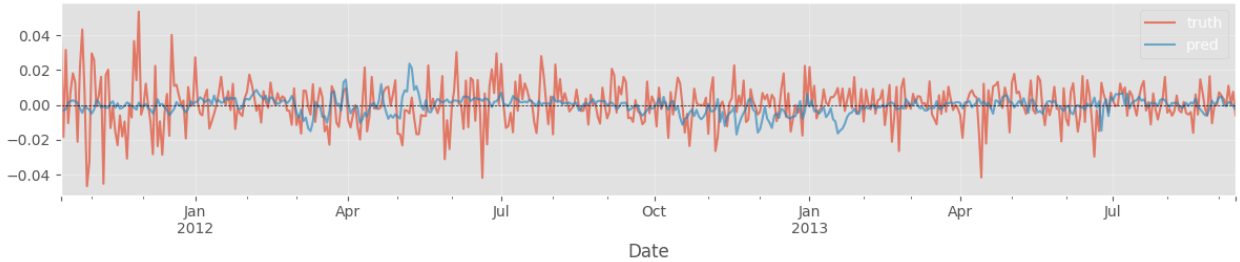

Figure 5: XLE Prediction vs Actual

6. Reflection & Conclusion

The development of this framework highlights the extreme difficulty of financial return prediction. While the hit ratios slightly exceed the 50% random benchmark, they demonstrate that even advanced DL architectures like VSN+LSTM face significant signal-to-noise challenges in liquid markets.

Successes: The modularity of the pipeline, from automated data retrieval to automated hyperparameter tuning, allows for rapid testing of new asset classes. The inclusion of the VSN layer provided a clear interpretability benefit by visualizing which macro factors (e.g., VIX, Credit Spreads) dominated the model's attention during different market regimes.

Future Improvements: Future iterations could integrate "position-sizing" optimization directly into the loss function, as suggested by the Oxford benchmark, rather than focusing solely on return prediction accuracy.

References

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1. Deep Learning for Financial Time Series: A Large-Scale Benchmark (Oxford Machine Learning Research Group).